Automated Car Parking System

Submitted as a Final Lab Project Report for the Course Robotics Lab
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Certificate of Approval

This is to certify that the project report entitled **"Automated Car Parking System"** has been prepared by **Md Shahid Al Mamim**, **Md Mizanur Rahman**, and **Lamia Akter Jesmin** as a final lab project report for the course **Robotics Lab** under the Department of Computer Science and Engineering, University of Asia Pacific. The report is accepted as a satisfactory submission for the course requirement.

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Abstract

Parking management is a major challenge in urban areas because manual systems often cause delay, confusion, and inefficient use of parking spaces. This project presents an **Automated Car Parking System** that integrates robotics and IoT technologies to automate vehicle entry, exit, and parking slot monitoring. The system uses Arduino UNO as the main controller, IR sensors for vehicle detection, RFID for authorized access, a 16x2 I2C LCD for displaying available slots, an ESP-01S Wi-Fi module for online data transmission, and a servo motor-controlled two-link robotic arm as the gate mechanism. When a vehicle is detected at the entrance, the system checks slot availability and requests RFID authentication. If the card is valid, the robotic arm gate opens, the slot count is updated, and the entry record is sent to the online server. During vehicle exit, the system detects the vehicle, opens the gate, updates the slot count, and stores the exit record. The prototype was tested under different entry and exit conditions, and the results showed successful vehicle detection, RFID verification, robotic gate operation, LCD update, and IoT data transmission. The proposed system reduces manual effort, improves parking efficiency, and demonstrates a practical application of robotics and IoT in smart parking management.

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Chapter 1

Introduction

In recent years, rapid urbanization and the increasing number of vehicles have made parking management a major challenge in residential, commercial, institutional, and public areas. Traditional parking systems usually depend on manual gate control, manual slot checking, and manual record keeping. This process is time-consuming, less accurate, and inefficient when the number of vehicles is high.

Automation, robotics, embedded systems, and Internet of Things (IoT) technologies can improve parking management by detecting vehicles, verifying access, controlling gates, updating available slots, and storing parking data automatically. In this project, an **Automated Car Parking System** is designed using Arduino UNO, IR sensors, RFID, LCD display, ESP-01S Wi-Fi communication, and a servo motor-controlled two-link robotic arm gate mechanism.

The system detects a vehicle at the entrance using an IR sensor and checks parking slot availability. If a slot is available, the user scans an RFID card for authentication. After successful verification, the Arduino controls the servo motor to move the two-link robotic arm gate. The available slot count is updated on the LCD and the entry record is sent to an online server. During exit, the system detects the vehicle, opens the robotic arm gate, increases the available slot count, and stores the exit record. Thus, the proposed system combines robotic gate control and IoT-based monitoring to create a low-cost smart parking prototype.

1.1 Background and Motivation

Parking space management has become difficult in busy places such as shopping malls, offices, universities, hospitals, and residential buildings. Drivers often waste time searching for available parking spaces, which increases congestion near parking areas. Manual systems may also create errors in slot counting and vehicle entry-exit records.

The motivation of this project is to develop a simple, low-cost, and automated parking system suitable for academic prototype implementation. The system uses IR sensors for vehicle detection, RFID for authorized access, an LCD for real-time slot display, ESP-01S for online data transmission, and a two-link robotic arm as the gate mechanism. The robotic arm adds a robotics-based control feature and demonstrates practical use of actuator movement in parking automation.

1.2 Problem Statement

Traditional parking systems have several limitations. Manual gate operation can cause delays during vehicle entry and exit. Manual slot counting can be inaccurate, and drivers may not know whether parking space is available before reaching the gate. Manual record keeping is also difficult to maintain for a long time.

Security is another concern in controlled parking areas. Without authentication, unauthorized vehicles may enter the parking area. Therefore, an automated system is needed that can detect vehicles, verify authorized users, control the gate, update slot availability, and store parking data using IoT communication.

1.3 Objectives of the Project

The main objective of this project is to design and implement an Automated Car Parking System that can control vehicle entry and exit using sensors, RFID, microcontroller-based control, IoT communication, and a two-link robotic arm gate.

The specific objectives are:

1. To design a prototype automated parking system using Arduino UNO.
2. To detect vehicles at entry and exit points using IR sensors.
3. To use RFID for authorized vehicle entry.
4. To control a servo motor-based two-link robotic arm gate.
5. To display available parking slots on a 16x2 I2C LCD.
6. To update parking slot count automatically after entry and exit.
7. To send entry and exit records to an online server using ESP-01S.
8. To demonstrate the integration of robotics and IoT in parking automation.

1.4 Research Questions

The project is guided by the following research questions:

1. How can an Arduino-based system automate vehicle entry and exit?
2. How effectively can IR sensors detect vehicle presence?
3. How can RFID be used for authorized parking access?
4. Can a servo-controlled two-link robotic arm operate as a reliable gate?
5. How can parking slot information be updated and stored automatically?

1.5 Scope of the Work

This project focuses on a prototype-level Automated Car Parking System. The system uses Arduino UNO, IR sensors, RFID module, ESP-01S Wi-Fi module, LCD display, and a servo motor-controlled two-link robotic arm. The main functions include vehicle detection, RFID authentication, robotic gate control, parking slot counting, LCD-based status display, and IoT-based data transmission.

The scope is limited to a small-scale lab prototype. It does not include automatic number plate recognition, camera-based detection, payment system, mobile reservation, GPS guidance, or large-scale commercial deployment. However, these features can be added in future versions.

1.6 Expected Outcomes

The expected outcome is a functional prototype that can detect vehicles, verify RFID cards, operate the two-link robotic arm gate, update slot availability, display parking status, and send parking records online. The system is expected to reduce manual effort, improve parking efficiency, and demonstrate the practical application of robotics and IoT in smart parking management.

1.7 Impacts of the Project

The proposed system can improve parking management by reducing manual gate operation and manual slot counting. It can save time for drivers, reduce congestion near parking gates, and support more organized parking control. Since the system stores parking data online, data privacy and secure communication should be considered in real deployment.

1.7.1 Impact on Societal and Cultural Issues

The system can support society by making parking more efficient and reducing waiting time. It also encourages the use of automation and IoT-based solutions in daily life, supporting smart city development and technology awareness.

1.7.2 Impact on Health, Safety, and Legal Issues

The automated gate operation can reduce human error and improve safety during vehicle entry and exit. The system has no direct health impact, but it can reduce driver stress by making parking faster. From a legal perspective, RFID and entry-exit data should be protected from unauthorized access.

1.7.3 Impact on Environment and Sustainability Issues

Efficient parking management can reduce unnecessary vehicle movement and waiting time, which may help lower fuel consumption and emissions. The system also uses low-cost and low-power electronic components suitable for sustainable prototype development.

1.8 Report Organization

The report is organized as follows:

Chapter 1: Introduction
This chapter presents the background, motivation, problem statement, objectives, scope, expected outcomes, impacts, and report organization.

Chapter 2: Related Works
This chapter reviews existing smart parking systems, IoT-based parking management, RFID access control, and automated gate systems.

Chapter 3: System Analysis and Design
This chapter describes requirements, system architecture, hardware setup, circuit connection, and robotic arm gate structure.

Chapter 4: Methodology and Implementation
This chapter explains the development method, tools, data handling, algorithm,

flowchart, and module implementation.

Chapter 5: Results and Evaluation

This chapter presents testing setup, performance metrics, result analysis, discussion, and limitations.

Chapter 6: Project Planning and Budget

This chapter includes the project timeline, cost estimation, and resource planning.

Chapter 7: Standards and Design Constraints

This chapter discusses standards, design constraints, CEP, CEA, and knowledge profile mapping.

Chapter 8: Conclusion and Future Work

This chapter summarizes the project achievements, limitations, and future improvement possibilities.

Chapter 2

Related Works

This chapter presents the background concepts, related works, and existing systems associated with the proposed **Automated Car Parking System**. It briefly explains the main technologies used in the project and identifies the limitations of existing parking solutions.

2.1 Preliminaries

The proposed system combines embedded systems, sensors, RFID authentication, robotic actuation, LCD display, and IoT-based data communication. These components work together to detect vehicles, verify users, control the gate, update parking slots, and store parking records online.

2.1.1 Internet of Things

The Internet of Things (IoT) connects physical devices to the internet for data exchange and remote monitoring [7]. In this project, IoT is used to send vehicle entry and exit records to an online server or database through the ESP-01S Wi-Fi module.

2.1.2 Embedded System and Arduino UNO

An embedded system is designed to perform a specific task using hardware and software. The Arduino UNO works as the main controller of this project. It receives input from IR sensors and the RFID module, processes the logic, controls the servo motor, updates the LCD, and communicates with the Wi-Fi module [1].

2.1.3 IR Sensor

IR sensors are used to detect vehicle presence at the entry and exit points. When a vehicle is detected, the sensor sends a digital signal to the Arduino, which starts the entry or exit process.

2.1.4 RFID Technology

RFID is used for contactless identification. In this project, the RFID reader scans a card UID and compares it with authorized IDs. If the card is valid, the system allows the vehicle to enter; otherwise, access is denied [10].

2.1.5 Servo Motor and Two-Link Robotic Arm

A servo motor provides controlled angular movement and is commonly used in robotics [2]. In this project, the servo motor controls a two-link robotic arm that works as the automatic gate mechanism. The robotic arm opens when access is granted and closes after the vehicle passes.

2.1.6 LCD Display and ESP-01S Wi-Fi Module

The 16x2 I2C LCD displays system messages and available parking slots. The ESP-01S Wi-Fi module sends parking event data to the online server using serial AT commands. This supports digital record keeping and remote monitoring.

2.1.7 Online Database

An online database stores entry and exit records so that parking information can be monitored remotely. Firebase Realtime Database is one example of a real-time cloud database that can store and synchronize data across connected clients [6].

2.2 Related Works / Existing Systems

Smart parking systems have been developed using different technologies such as sensors, IoT platforms, RFID authentication, cloud databases, and mobile applications. These systems aim to reduce parking time, improve space utilization, and support smart city applications.

2.2.1 Sensor-Based Smart Parking Systems

Sensor-based systems use IR, ultrasonic, magnetic, or camera-based sensors to detect vehicle presence. Choudhary *et al.* proposed an IoT-based smart parking system using sensors and microcontrollers to identify available parking spaces [3]. Such systems are simple and low-cost, but many of them focus only on slot detection and do not include authentication or robotic gate control.

2.2.2 IoT-Based Parking Systems

IoT-based parking systems connect hardware devices with online platforms for remote monitoring. Elsonbaty and Shams proposed a smart parking management system using Arduino, Android application, and IoT technology [4]. Mamun *et al.* also developed an IoT-enabled smart parking system using integrated sensors and mobile applications [9]. These systems improve monitoring, but many of them do not include a robotics-based gate mechanism.

2.2.3 RFID-Based Access Control Systems

RFID-based systems are widely used for controlled parking areas such as offices, universities, and residential buildings. They allow entry only for authorized users. However, some RFID-based systems do not include complete parking slot monitoring or online data storage. The proposed system combines RFID authentication with slot counting, LCD display, and IoT-based record storage.

2.2.4 Cloud-Based and Middleware Parking Systems

Cloud-based systems store parking information on remote servers. Ji *et al.* proposed a cloud-based car parking middleware for IoT-based smart cities [8]. Middleware can improve data handling and scalability, but it may increase complexity. The proposed system uses a lightweight online communication approach suitable for a prototype.

2.2.5 Automated Gate Control Systems

Automated gates are commonly controlled using DC motors, stepper motors, or servo motors. Most basic parking projects use a simple servo barrier gate. In this project, a two-link robotic arm gate is used instead, which adds a robotics-based mechanical movement to the system.

2.3 Comparative Analysis of Related Works

Existing systems usually focus on one or two major features such as slot detection, IoT monitoring, RFID access, or gate automation. The proposed system combines these features with a two-link robotic arm gate, making it suitable for both smart parking and robotics demonstration.

Table 2.1: Comparison of Related Parking System Approaches

Approach / Study	**Technology**	**Strengths**	**Limitations**	**Relation with Proposed System**
Sensor-based parking [3]	Sensors, microcontroller	Low cost and simple	Limited access control	Proposed system adds RFID, IoT, and robotic arm gate.
IoT parking system [4, 9]	Arduino, IoT, mobile app	Remote monitoring	More software dependency	Proposed system focuses on compact prototype automation.
Cloud parking middleware [8]	Cloud and middleware	Better data management	More complex architecture	Proposed system uses lightweight online data transfer.
RFID access control	RFID reader and card	Improves security	Limited slot monitoring	Proposed system combines RFID with slot update and IoT.
Servo gate system	Servo and sensor	Simple automation	Limited robotic motion	Proposed system uses a two-link robotic arm gate.

2.4 Research Gap / Limitations of Existing Methods

Many existing smart parking systems have limitations. Some systems only detect available slots but do not automate user authentication or gate operation. Some systems use simple servo barriers that do not show significant robotic movement. Other systems provide remote monitoring but require complex mobile applications or cloud architecture.

Another common limitation is the lack of complete integration. A system may include sensors but not RFID authentication, or it may include RFID but not IoT-based data storage. The proposed Automated Car Parking System addresses these gaps by combining vehicle detection, RFID-based access control, two-link robotic arm gate operation, LCD-based slot display, and IoT-based parking record storage in one low-cost prototype.

2.5 Summary

This chapter discussed the basic technologies and related works associated with the Automated Car Parking System. Existing systems show the usefulness of sensors, IoT, RFID, cloud storage, and automated gates in parking management. However, many systems lack a combined approach with robotic gate movement. The proposed system addresses this gap by integrating IR sensors, RFID authentication, Arduino control, ESP-01S communication, LCD display, and a two-link robotic arm gate for smart parking automation.

Chapter 3

System Analysis and Design

This chapter presents the system analysis and design of the proposed **Automated Car Parking System**. It covers the main requirements, system architecture, embedded system diagram, hardware setup, circuit connection, robotic arm gate structure, and design rationale.

3.1 Requirement Analysis

Requirement analysis defines what the system should perform and how it should behave. The requirements were identified by observing common parking problems, reviewing existing smart parking systems, and discussing the expected system behavior among group members. The main stakeholders are drivers, parking administrators, and system maintainers.

The system must detect vehicles, verify authorized users, operate the robotic arm gate, update parking slot information, display real-time status, and send parking records to an online server or database. The requirements are divided into functional and non-functional requirements.

3.1.1 Functional Requirements

Table 3.1: Functional Requirements

ID	Requirement Description
FR1	Detect vehicles at the entrance using an IR sensor.
FR2	Detect vehicles at the exit using an IR sensor.
FR3	Verify authorized users using RFID cards.
FR4	Open and close the two-link robotic arm gate using a servo motor.
FR5	Update available parking slot count after valid entry and exit.
FR6	Display current slot availability on a 16x2 I2C LCD.
FR7	Send entry and exit records to an online server using Wi-Fi.
FR8	Prevent vehicle entry when no parking slot is available.

3.1.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

ID	Requirement Description
NFR1	The system should respond quickly after vehicle detection.
NFR2	The robotic arm gate should move smoothly and safely.
NFR3	The system should be simple and user-friendly.
NFR4	The design should be low-cost and suitable for prototype implementation.
NFR5	The slot count should remain accurate during entry and exit.
NFR6	Online records should be transmitted reliably when Wi-Fi is available.

3.2 System Overview / Architecture

The Automated Car Parking System is an embedded and IoT-based system. It uses IR sensors for vehicle detection, RFID for authentication, Arduino UNO for processing, a servo motor-controlled two-link robotic arm for gate movement, LCD for local display, and ESP-01S Wi-Fi module for online data transmission.

During entry, the system detects a vehicle, checks slot availability, verifies RFID, opens the robotic arm gate, updates the slot count, and sends the entry record online. During exit, the system detects the vehicle, opens the gate, updates the slot count, and sends the exit record.

3.2.1 High-Level Architecture

Fig. 3.1 shows the high-level workflow of the system. The Arduino UNO acts as the central controller and communicates with the sensors, RFID module, LCD, servo motor, and Wi-Fi module.

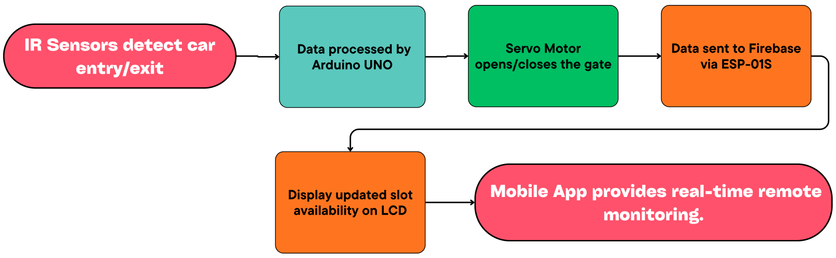

Figure 3.1: High-Level Workflow of Automated Car Parking System

3.2.2 Embedded System Diagram

The embedded system diagram in Fig. 3.2 shows the interaction among the hardware and software modules. The Arduino UNO receives input from the entry IR sensor, exit IR sensor, and RFID reader. It then controls the servo motor-based two-link robotic arm gate, updates the LCD, and communicates with the ESP-01S Wi-Fi module.

The ESP-01S sends parking event data to the middleware server, which stores the information in the online database. The stored data can later be monitored through an online interface or mobile application.

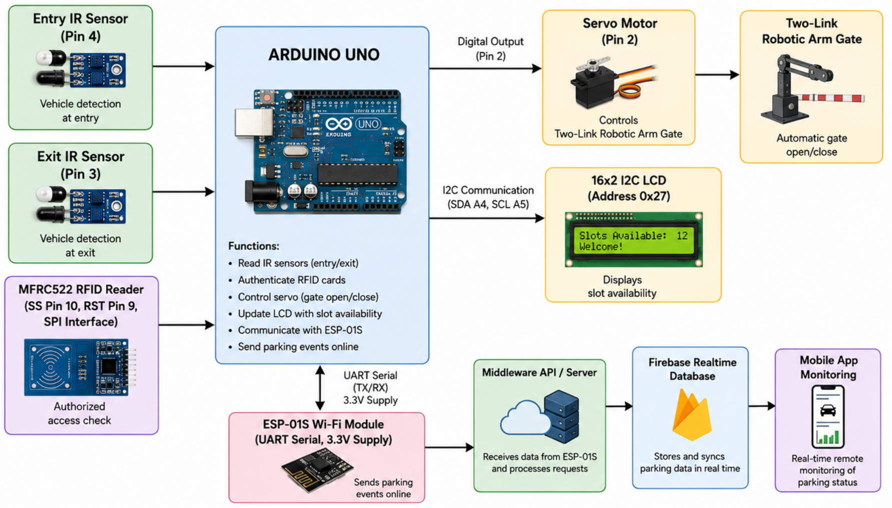

Figure 3.2: Embedded System Diagram of Automated Car Parking System

3.3 Hardware Setup

The hardware setup includes the main electronic and mechanical components required for vehicle detection, authentication, gate control, display, and communication.

Table 3.3: Main Hardware Components

Component	Function
Arduino UNO	Main controller of the system.
IR Sensors	Detect vehicle presence at entry and exit points.
RFID Module	Reads RFID cards for user authentication.
Servo Motor	Controls the two-link robotic arm gate.
Two-Link Robotic Arm	Works as the automatic gate mechanism.
16x2 I2C LCD	Displays slot availability and system messages.
ESP-01S Wi-Fi Module	Sends parking data to the online server or database.
Power Supply	Provides required power to the circuit and components.

3.3.1 Circuit Connection

The circuit connects all modules with the Arduino UNO. The IR sensors are connected to digital pins for vehicle detection. The RFID module uses SPI communication. The LCD uses I2C communication, which reduces wiring complexity. The servo motor receives control signals from the Arduino, and the ESP-01S communicates through serial AT commands for internet connectivity.

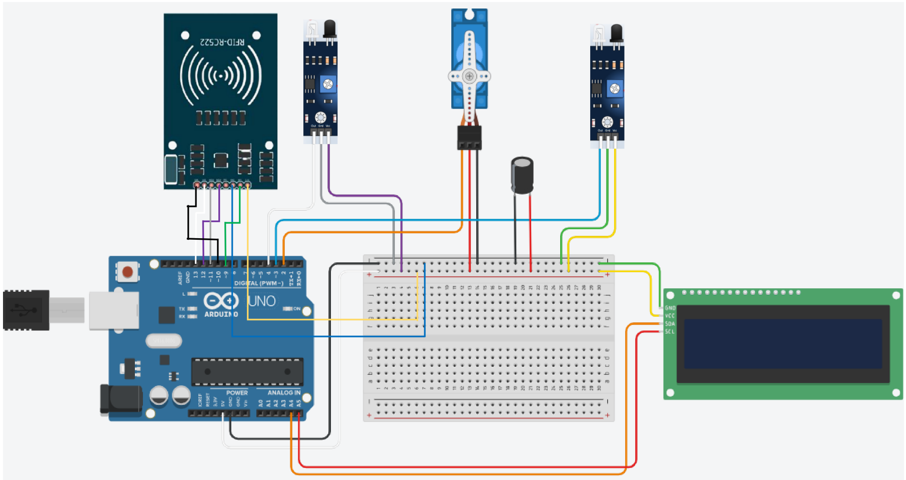

Figure 3.3: Circuit Connection of the Proposed System

3.3.2 Robot Structure Details

The gate mechanism is designed using a servo motor-controlled two-link robotic arm. The two-link structure provides controlled robotic movement instead of a simple static barrier. When the system grants permission, the servo motor moves the arm to the open position. After the vehicle passes, the arm returns to the closed position.

This structure makes the system suitable for a robotics lab project because it demonstrates actuator control, mechanical movement, and automated response based on sensor input.

3.4 Design Alternatives and Rationale

Several alternatives were considered during design. A manual gate was rejected because it does not provide automation. A simple servo barrier was easier to build but did not demonstrate enough robotic motion. A camera-based system was also considered but would increase cost and complexity.

The final design was selected because it balances cost, functionality, and simplicity. Arduino UNO, IR sensors, RFID, LCD, ESP-01S, and a two-link robotic arm provide a practical solution for a low-cost robotics and IoT-based parking prototype. The design also allows future improvements such as mobile app integration, camera-based vehicle recognition, or payment support.

3.5 Summary

This chapter described the system analysis and design of the Automated Car Parking System. The functional and non-functional requirements, system architecture, embedded system diagram, hardware setup, circuit connection, robotic arm gate mechanism, and design rationale were presented. The selected design is simple, low-cost, and suitable for demonstrating robotics and IoT-based parking automation.

Chapter 4

Methodology and Implementation

This chapter describes the methodology and implementation process of the **Automated Car Parking System**. It explains the development steps, tools, data handling, algorithm, flowchart, and main implementation modules.

4.1 Methodological Framework

The project was developed using a step-by-step embedded system approach. First, the parking problem and required features were identified. Then suitable components were selected based on cost, availability, and compatibility with Arduino UNO. After that, the circuit and robotic arm gate structure were designed. Each module was tested separately and finally integrated into one complete system.

The methodology followed in this project is given below:

1. Identify the parking management problem and define objectives.
2. Select required hardware components.
3. Design the circuit and two-link robotic arm gate.
4. Develop Arduino code for sensors, RFID, gate control, LCD, and Wi-Fi communication.
5. Test each module individually.
6. Integrate all modules into the final prototype.
7. Test the system under entry and exit conditions.

4.2 Tools, Libraries, and Technologies Used

The system was developed using both hardware and software tools. Arduino UNO was used as the main controller, and Arduino IDE was used for writing and uploading the program.

Table 4.1: Tools, Libraries, and Technologies Used

Tool / Technology	Purpose
Arduino IDE	Writing, compiling, and uploading Arduino code.
Arduino UNO	Main microcontroller of the system.
C/C++	Programming language for Arduino.
IR Sensors	Vehicle detection at entry and exit points.
MFRC522 RFID Module	RFID card-based authentication.
Servo Motor	Controls the two-link robotic arm gate.
16x2 I2C LCD	Displays slot availability and system messages.
ESP-01S Wi-Fi Module	Sends parking data to the online server.
Firebase / Online Database	Stores vehicle entry and exit records.

The main Arduino libraries used are `Wire.h`, `LiquidCrystal_I2C.h`, `Servo.h`, `SPI.h`, and `MFRC522.h`. These libraries support I2C communication, LCD control, servo movement, SPI communication, and RFID reading.

4.3 Data Collection and Processing

This project does not use a machine learning dataset. It uses real-time data from hardware modules. The main data includes IR sensor status, RFID UID, slot count, event type, timestamp, and owner information.

Table 4.2: System Data Description

Data Type	Description
IR Sensor Data	Detects vehicle presence at entry or exit.
RFID UID	Stores the unique ID of the scanned RFID card.
Slot Count	Maintains available parking slots.
Event Type	Identifies entry or exit operation.
Timestamp	Stores the time of the parking event.
Owner Information	Identifies the user linked with the RFID card.

The Arduino processes the collected data. RFID UID is converted into a readable string, trimmed, and matched with authorized UID values. The system also prepares a timestamp and checks the slot count to avoid invalid updates. For every valid entry or exit, a JSON packet is created and sent online through the ESP-01S Wi-Fi module.

4.4 Model Development / Algorithm Design

The project does not use a machine learning model. It follows rule-based control logic. The Arduino continuously checks the entry and exit IR sensors. If a vehicle is detected at the entry point, the system checks slot availability and requests RFID authentication. If the RFID card is valid, the gate opens and the slot count is updated.

4.4.1 System Flowchart

Fig. 4.1 shows the working flow of the Automated Car Parking System. It includes hardware initialization, vehicle detection, slot checking, RFID verification, robotic arm gate control, LCD update, and online data transmission.

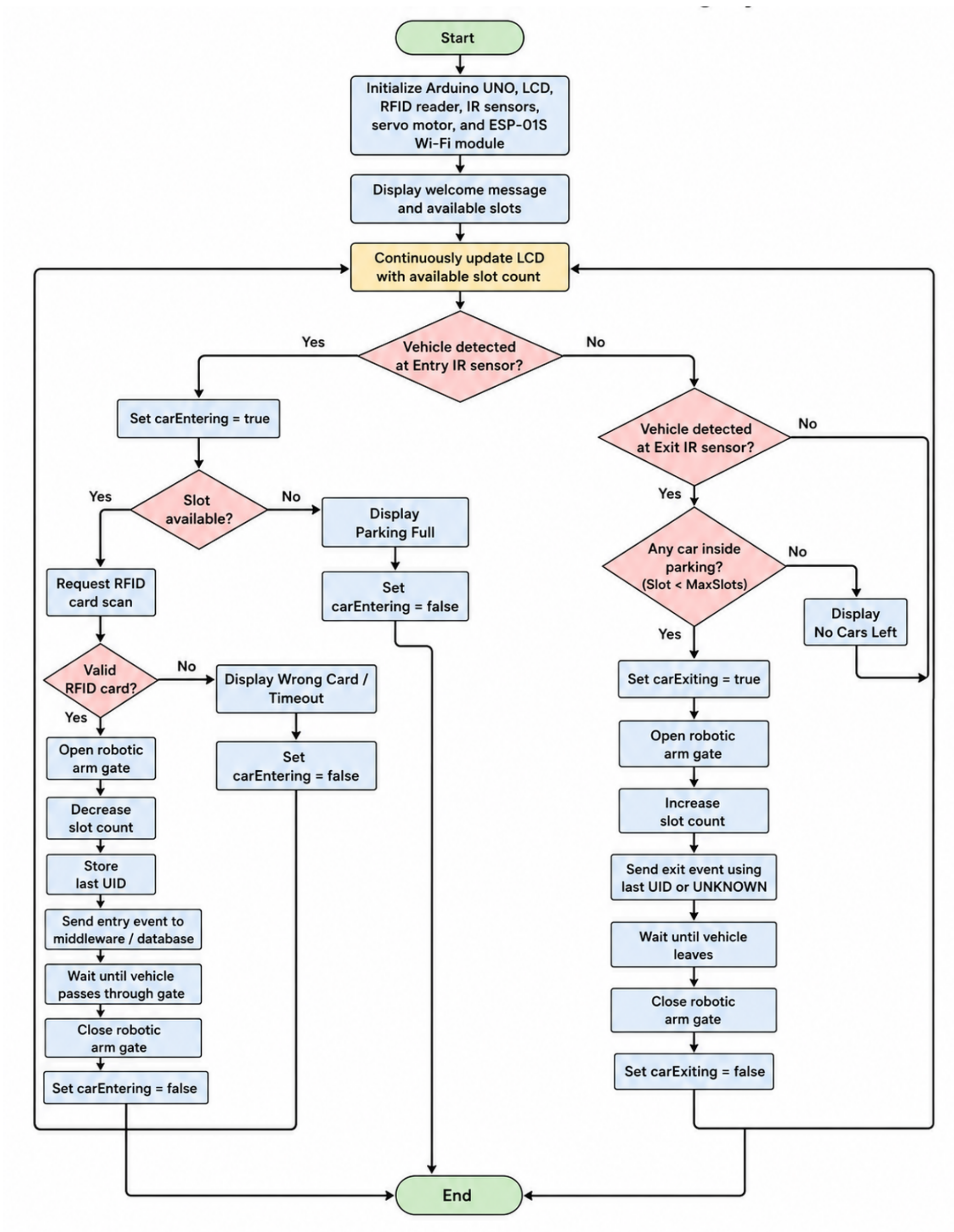

Figure 4.1: Flowchart of Automated Car Parking System

The simplified algorithm is given below:

Algorithm 1 System Operation Algorithm

```
Initialize LCD, RFID, servo motor, IR sensors, and Wi-Fi module
Set maximum parking slot value
while system is active do
    if entry sensor detects a car then
        if slot is available then
            Request RFID card scan
            if RFID card is valid then
                Open robotic arm gate
                Decrease slot count
                Send entry data to online server
                Close robotic arm gate
            else
                Display access denied message
            end if
        else
            Display parking full message
        end if
    end if
    if exit sensor detects a car then
        Open robotic arm gate
        Increase slot count
        Send exit data to online server
        Close robotic arm gate
    end if
    Update LCD display
end while
```

4.5 System Implementation / Modules Description

The system was implemented by dividing it into functional modules. Each module performs a specific task and communicates with the Arduino UNO.

4.5.1 Vehicle Detection Module

Two IR sensors are used to detect vehicles at the entry and exit points. When a vehicle is detected, the sensor sends a digital signal to the Arduino and starts the entry or exit process.

4.5.2 RFID Authentication Module

The RFID module checks whether the user is authorized. If the scanned card matches a stored UID, the system allows entry. Otherwise, the LCD displays a wrong card or access denied message.

4.5.3 Robotic Arm Gate Module

The gate is controlled by a servo motor connected to a two-link robotic arm. When entry or exit is allowed, the servo moves the arm to the open position. After the vehicle passes, the arm returns to the closed position. Smooth movement is created by changing the servo angle step by step.

4.5.4 LCD Display Module

The 16x2 I2C LCD displays welcome messages, RFID scanning messages, wrong card alerts, parking full messages, gate status, and available slot count. This helps users understand the current system condition.

4.5.5 IoT Communication Module

The ESP-01S Wi-Fi module sends parking data to the online server using AT commands. For each valid event, the Arduino creates a JSON message and sends it through an HTTP POST request.


```
{
  "uid": "F6 A9 23 03",
  "event": "entry",
  "owner": "MRJ",
  "timestamp": "00:05:23"
}
```

4.6 Hardware and Software Requirements

Table 4.3: Hardware and Software Requirements

Category	**Requirements**
Controller	Arduino UNO
Sensors	Two IR sensors for entry and exit detection
Authentication	MFRC522 RFID reader and RFID cards
Actuator	Servo motor with two-link robotic arm mechanism
Display	16x2 I2C LCD display
Communication	ESP-01S Wi-Fi module
Power Supply	5V supply for Arduino and connected modules
Programming Tool	Arduino IDE
Programming Language	C/C++ for Arduino
Libraries	Wire, LiquidCrystal_I2C, Servo, SPI, MFRC522
Online Platform	Online server or database for parking data storage

4.7 Summary

This chapter explained the methodology and implementation of the Automated Car Parking System. It described the development steps, tools, system data, processing method, flowchart, algorithm, implementation modules, and hardware-software requirements. The final implementation combines vehicle detection, RFID authentication, robotic arm gate control, LCD display, and IoT-based data transmission.

Chapter 5

Results and Evaluation

This chapter presents the results and evaluation of the **Automated Car Parking System**. The system was tested to evaluate vehicle detection, RFID authentication, robotic arm gate movement, parking slot update, LCD output, and IoT data transmission.

5.1 Experimental Setup

The prototype was tested in a lab environment using a small-scale parking model. The setup included Arduino UNO, two IR sensors, MFRC522 RFID module, servo motor, two-link robotic arm gate, 16x2 I2C LCD, and ESP-01S Wi-Fi module. Toy cars or hand-based object movement were used to simulate vehicle entry and exit.

The entry IR sensor was placed near the entrance gate, and the exit IR sensor was placed near the exit point. The RFID reader was placed at the entrance for authorized access. The servo motor controlled the robotic arm gate, the LCD displayed available slots, and the Wi-Fi module sent parking records to the online server.

Table 5.1: Experimental Setup

Item	**Description**
Controller	Arduino UNO
Vehicle Detection	IR sensors at entry and exit points
Authentication	MFRC522 RFID reader and RFID cards
Gate Mechanism	Servo motor-controlled two-link robotic arm
Display Unit	16x2 I2C LCD display
Communication	ESP-01S Wi-Fi module
Parking Capacity	4 prototype parking slots
Testing Method	Repeated entry and exit simulation

5.2 Performance Evaluation

The system was evaluated based on functional correctness, response behavior, and module reliability. The main testing areas were vehicle detection, RFID verification, gate operation, slot count update, LCD display, and IoT data transmission.

Table 5.2: Functional Test Result

Test Case	Total Test	Successful	Result
Entry IR detection	20	20	Successful
Exit IR detection	20	20	Successful
Valid RFID detection	15	15	Successful
Invalid RFID rejection	10	10	Successful
Robotic arm gate operation	20	19	Mostly successful
Parking slot update	20	20	Successful
LCD status display	20	20	Successful
IoT data transmission	20	18	Mostly successful

The test results show that the IR sensors detected vehicles accurately during prototype testing. RFID authentication worked correctly for both valid and invalid cards. The robotic arm gate opened and closed successfully in most cases, although minor movement variation occurred due to servo positioning and mechanical alignment. The LCD displayed correct slot information, and IoT data transmission was successful in most tests, depending on Wi-Fi stability.

Table 5.3: Overall Performance Summary

Performance Area	Observed Performance	Comment
Vehicle Detection	100%	Accurate in lab testing
RFID Authentication	100%	Valid and invalid cards detected correctly
Robotic Arm Operation	95%	Minor adjustment needed
Slot Count Update	100%	Updated correctly
LCD Display	100%	Displayed correct status
IoT Data Transmission	90%	Depended on Wi-Fi connection

5.3 Discussion on Findings

The experimental results indicate that the main objectives of the project were achieved. The system successfully detected vehicle presence, verified RFID cards, controlled the two-link robotic arm gate, updated parking slot count, displayed information on the LCD, and sent data to the online server.

Compared with a manual parking system, the proposed system reduces human involvement in gate control, slot counting, and record keeping. Compared with a simple servo gate system, this project adds RFID authentication, IoT-based data storage, and a two-link robotic arm mechanism, making it more suitable for robotics lab demonstration.

The IR sensors performed well at short range, and RFID provided a simple and effective access control method. The servo motor controlled the robotic arm properly, but mechanical alignment was important for smooth movement. The IoT module added remote record storage, although its performance depended on Wi-Fi connection quality and ESP-01S response time.

5.4 Limitations

Although the prototype performed successfully, it has some limitations. The system was tested on a small-scale model, so real-world implementation would require a stronger motor, better mechanical structure, and more durable materials. IR sensors may be affected by object position, distance, or lighting conditions. The robotic arm also needs proper alignment for stable movement.

The Wi-Fi communication depends on network availability. If the connection is weak, online data transmission may fail. The current prototype uses RFID only and does not include automatic number plate recognition, mobile application, payment system, camera-based detection, or large-scale database management.

5.5 Summary

This chapter presented the experimental setup, functional test results, performance summary, discussion, and limitations of the Automated Car Parking System. The results show that the system can perform vehicle detection, RFID authentication, robotic arm gate control, slot count update, LCD display, and IoT data transmission successfully. Overall, the project demonstrates a practical prototype of a robotics and IoT-based automated parking system.

Chapter 6

Project Planning and Budget

This chapter presents the project planning and budget estimation of the **Automated Car Parking System**. It shows the project timeline, required resources, and estimated cost for completing the prototype.

6.1 Project Timeline

The project was completed through requirement analysis, component selection, system design, hardware setup, coding, testing, integration, documentation, and presentation preparation. The eight-week timeline helped the team divide tasks properly and complete the project in an organized way.

Table 6.1: Eight-Week Project Timeline

Project Phase	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**
Requirement Analysis								
Component Selection								
System Design								
Circuit and Hardware Setup								
Arduino Coding								
Module Testing								
Final Integration								
Testing and Validation								
Report Writing and Presentation								

The project followed the planned timeline closely. Extra attention was needed during hardware integration because the two-link robotic arm gate required proper mechanical alignment for smooth movement.

6.2 Budget Estimation

The budget was estimated based on the required hardware components and supporting materials. Most software tools used in this project were free.

Table 6.2: Estimated Project Budget

Item / Resource	**Qty.**	**Unit Cost**	**Total Cost**
Arduino UNO	1	700	700
MFRC522 RFID Module with Card	1 Set	250	250
IR Sensors	2	100	200
Servo Motor	1	250	250
ESP-01S Wi-Fi Module	1	300	300
16x2 I2C LCD Display	1	350	350
Breadboard and Jumper Wires	1 Set	250	250
Robotic Arm / Gate Materials	1 Set	400	400
Power Supply and Connectors	1 Set	300	300
Printing and Presentation Materials	-	300	300
Total Estimated Budget			**3,300 BDT**

The total estimated cost of the project was approximately **3,300 BDT**. The cost may vary depending on component quality and market availability.

6.3 Software and Service Cost

The software tools and online services were used in free or prototype-accessible versions.

Table 6.3: Software and Service Cost

Software / Service	**Purpose**	**Cost**
Arduino IDE	Code development and upload	Free
Arduino Libraries	LCD, RFID, Servo, and SPI support	Free
Online Database / Server	Parking data storage	Free / Prototype Use
GitHub Repository	Code and report submission	Free

6.4 Summary

This chapter presented the eight-week project timeline and estimated budget of the Automated Car Parking System. The project was completed through planning, hardware setup, coding, testing, integration, and documentation phases. The budget shows that the system can be developed using low-cost hardware and free software tools, making it feasible for academic prototype implementation.

Chapter 7

Standards and Design Constraints

This chapter discusses how the **Automated Car Parking System** satisfies engineering standards, design constraints, Complex Engineering Problem (CEP) attributes, Complex Engineering Activity (CEA) attributes, and Knowledge Profile requirements. Since the project combines hardware, software, robotics, and IoT communication, safety, reliability, privacy, sustainability, and professional responsibility were considered during design and implementation.

7.1 Compliance with Standards and Professional Practice

The project followed basic engineering practices for embedded system design, hardware interfacing, modular programming, IoT communication, documentation, and testing. Relevant documentation such as Arduino UNO, Servo library, MFRC522 RFID module, ESP8266, and Firebase documentation was considered during development [1, 2, 5, 6, 10].

7.1.1 Software and Hardware Practice

The software was developed using Arduino C/C++ in Arduino IDE. The code was organized into separate functions for RFID reading, gate control, LCD update, Wi-Fi connection, timestamp generation, and data transmission. Standard libraries such as `Servo.h`, `SPI.h`, `MFRC522.h`, `Wire.h`, and `LiquidCrystal_I2C.h` were used.

The hardware design followed basic low-voltage embedded system interfacing practice. Arduino UNO was used as the controller, IR sensors were used for detection, MFRC522 used SPI communication, LCD used I2C communication, ESP-01S used serial communication, and the servo motor controlled the two-link robotic arm. Safe testing was performed using a small-scale prototype instead of a real vehicle gate.

7.1.2 Professional Considerations

The system stores RFID UID, entry/exit event, owner information, and timestamp. Therefore, privacy and responsible data use were considered. In real deployment, Wi-Fi credentials, database keys, and user records must be protected and should not be published in source code or reports.

7.2 Design Constraints

The project was affected by cost, component availability, sensor accuracy, power stability, mechanical alignment, and Wi-Fi reliability. Since it was developed as an academic prototype, low-cost components were selected while keeping the main functions operational.

7.2.1 Ethical, Safety, and Sustainability Considerations

The testing results and limitations were reported honestly. Related works and technical documentation were cited properly. RFID and online data should be handled responsibly to protect user privacy.

For safety, low-voltage components were used and the robotic arm was tested in a controlled lab environment. The servo movement was kept smooth to reduce sudden motion, and direct contact with the moving arm was avoided during operation.

From an environmental perspective, the system uses low-power and reusable electronic components. In practical use, automated parking can reduce unnecessary waiting time and vehicle movement near the gate, which may help reduce fuel consumption and emissions.

7.3 Complex Engineering Problem (CEP) Coverage

The project addresses a complex engineering problem because it integrates embedded control, sensors, authentication, robotic actuation, IoT communication, and real-time decision making.

Table 7.1: Mapping with Complex Engineering Problems (Ps)

Ps	Attributes	How Ps are addressed	Evidence
P1	Depth of Knowledge	Applies embedded systems, sensors, RFID, servo control, IoT, and database communication.	Ch. 2, 3, 4
P2	Conflicting Requirements	Balances low cost, real-time response, reliable detection, smooth gate movement, and online data transfer.	Ch. 3, 5
P3	Depth of Analysis	Analyzes sensor response, RFID validation, slot counting, robotic arm motion, and Wi-Fi transmission.	Ch. 4, 5
P4	Familiarity of Issues	Parking automation is known, but robotic arm gate with IoT monitoring creates project-specific challenges.	Ch. 1, 2, 3
P5	Applicable Standards	Uses standard Arduino libraries, hardware interfacing practices, and IoT data handling principles.	Sec. 7.1
P6	Stakeholder Needs	Supports drivers, parking managers, and maintainers through automation and monitoring.	Ch. 3, Appendix
P7	Interdependence	Sensors, RFID, servo, LCD, Wi-Fi, and server are interconnected; failure of one module affects the system.	Ch. 3, 4, 5

7.4 Complex Engineering Activities (CEA) Coverage

The project involves complex engineering activities because it required multiple resources, hardware-software interaction, implementation, testing, and evaluation.

Table 7.2: Mapping of Complex Engineering Activities (As)

As	Attributes	How As are addressed	Evidence
A1	Range of Resources	Uses Arduino UNO, IR sensors, RFID, ESP-01S, LCD, servo motor, robotic arm materials, libraries, and online database.	Ch. 3, 4
A2	Level of Interaction	Multiple modules interact, including sensors, RFID, servo, LCD, Wi-Fi module, and server.	Ch. 3, 4
A3	Innovation	Uses a two-link robotic arm gate instead of a simple static barrier.	Ch. 3, 5
A4	Society and Environment	Can reduce waiting time, improve parking management, reduce manual effort, and support smart city applications.	Ch. 1, 7
A5	Familiarity	Combines known technologies into a practical robotics-IoT parking automation prototype.	Ch. 2, 4

7.5 Knowledge Profile Mapping

Table 7.3: Knowledge Profile Mapping

Profile	**Knowledge Area**	**Project Evidence**
K1	Natural science and engineering fundamentals	Sensor detection, electrical interfacing, and servo control.
K2	Mathematics and computing	Control logic, slot counting, timestamp generation, and data formatting.
K3	Engineering fundamentals	Embedded system design using Arduino and electronic modules.
K4	Specialist knowledge	RFID authentication, IoT communication, and robotic arm gate control.
K5	Engineering design	System architecture, circuit design, and hardware-software integration.
K6	Engineering practice	Prototype development, testing, debugging, and documentation.
K7	Social, environmental, and safety knowledge	Parking efficiency, safety, data privacy, and sustainability considerations.
K8	Research literature and standards	Related works review and use of technical documentation.

7.6 Summary

This chapter presented the standards, design constraints, CEP coverage, CEA coverage, and Knowledge Profile mapping of the Automated Car Parking System. The project satisfies engineering requirements by integrating embedded control, robotics, IoT communication, and real-time parking management while considering safety, privacy, sustainability, and professional responsibility.

Chapter 8

Conclusion and Future Work

This chapter presents the conclusion of the **Automated Car Parking System** project. It summarizes the completed work, key findings, limitations, and possible future improvements.

8.1 Summary of the Work

The main objective of this project was to design and implement an Automated Car Parking System using robotics and IoT technologies. The system was developed using Arduino UNO, IR sensors, RFID module, ESP-01S Wi-Fi module, 16x2 I2C LCD display, servo motor, and a two-link robotic arm gate mechanism.

The system detects vehicles at the entry and exit points using IR sensors. During entry, it checks slot availability and verifies the RFID card before opening the gate. If the RFID card is valid and a slot is available, the servo motor moves the two-link robotic arm to open the gate. After vehicle entry, the slot count is updated on the LCD and the entry record is sent to the online server. During exit, the system opens the gate, updates the slot count, and sends the exit record online.

The final prototype was tested in a lab environment and performed successfully in vehicle detection, RFID authentication, robotic arm gate operation, slot count update, LCD display, and IoT-based data transmission.

8.2 Key Findings and Contributions

The project achieved its major objectives and produced a functional prototype. The key findings and contributions are listed below:

- IR sensors successfully detected vehicles at entry and exit points.
- RFID authentication allowed only authorized users to enter.

- The servo motor-controlled two-link robotic arm worked as an automatic gate.
- The LCD displayed real-time parking slot availability and system messages.
- The ESP-01S Wi-Fi module transmitted entry and exit records online.
- The system reduced manual gate control, slot counting, and record keeping.
- The project demonstrated practical integration of robotics and IoT in parking automation.

The main technical contribution of the project is the use of a two-link robotic arm gate instead of a simple static barrier. This makes the system more suitable for a robotics-based automation project.

8.3 Limitations of the Study

Although the system worked successfully as a prototype, it has some limitations. The system was tested on a small-scale model, so real-world implementation would require stronger motors, durable materials, and improved mechanical structure. IR sensors may be affected by object position, distance, and surrounding conditions. The robotic arm also requires proper alignment for stable movement.

The Wi-Fi communication depends on network availability, so weak internet connection may delay or stop online data transmission. The current system uses RFID-based authentication only and does not include automatic number plate recognition, mobile application, payment system, reservation system, or camera-based detection.

8.4 Recommendations and Future Work

The prototype can be improved in several ways. Future improvements may include:

- Adding automatic number plate recognition using a camera.
- Developing a mobile application for users and administrators.
- Adding online parking reservation and payment features.
- Improving data security for RFID records and online communication.
- Using stronger motors and durable materials for real gate operation.
- Adding backup power support for continuous operation.

- Improving the robotic arm design for smoother movement.
- Adding individual slot sensors for multiple parking spaces.

In conclusion, the Automated Car Parking System successfully fulfills the main goal of automating parking entry and exit using robotics and IoT technologies. The project provides a practical prototype that can be further improved for academic, residential, and commercial parking environments.

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Appendices

A Source Code / GitHub Link

The complete source code, final report PDF, diagrams, and relevant resources of the **Automated Car Parking System** will be uploaded to a GitHub repository for documentation and replication purposes.

GitHub Repository Link:

https://github.com/Mamim57/Automated-Car-Parking-System

The main Arduino source code file is:

- Robo_Car_Parking.ino

The source code includes IR sensor reading, RFID authentication, LCD update, servo motor-based robotic arm gate control, parking slot counting, and ESP-01S Wi-Fi based data transmission.

For security reasons, sensitive information such as Wi-Fi SSID, Wi-Fi password, API keys, and database credentials should not be published directly. These values should be replaced with placeholders before public submission.

```
String ssid = "YOUR_WIFI_NAME";
String password = "YOUR_WIFI_PASSWORD";
```

B Team Responsibilities

Table 1: Team Member Responsibilities

Team Member	**Contribution**
Md. Mizanur Rahman	Arduino coding, system logic, Firebase/online database setup, Wi-Fi data communication, and software debugging.
Md. Shahid Al Mamim	Hardware setup, component selection, circuit connection, robotic arm gate setup, system testing, and report writing.
Lamia Akter Jesmin	Survey preparation, system analysis, requirement analysis, report writing, and Wi-Fi module integration support.

C Ethics and Compliance Statements

The project was developed for academic and prototype demonstration purposes. The survey data was used only for academic research, and personal information was not shared in the report. RFID and parking event data were used only for system testing.

The following ethical issues were considered:

- Testing results were reported honestly.
- Related works and technical documents were cited properly.
- Wi-Fi passwords, API keys, and database credentials were kept private.
- The robotic arm gate was tested safely in a controlled lab environment.
- RFID and parking event data were used only for project demonstration.

In a real deployment, proper data privacy, secure communication, and user consent should be maintained.

D Survey 1: Initial User Need Analysis

The first survey was conducted to understand user perception about the product concept and its possible usefulness. During the initial concept discussion, both automated car parking and an expression bot idea were mentioned. However, the final implemented project focused only on the **Automated Car Parking System**. Therefore, the expression bot was treated as an early brainstorming idea and was not included in the final implementation.

Survey Questions

1. What is your job role?
2. Are you involved in purchasing or decision making?
3. Have you understood the product concept?
4. Do you think the product will be helpful for your workplace?
5. Do you have any additional comments?

Survey Response Summary

Table 2: Summary of Survey 1 Response

Question / Area	**Response Summary**
Respondent Role	Assistant Coordinator in Web Service and Front-end Software Development.
Decision Making Involvement	Yes.
Understanding of Concept	Rated 3 out of 5.
Workplace Helpfulness	Rated 3 out of 5.
Additional Comment	The respondent found the concept interesting and mentioned that automated car parking has practical value if implemented properly. The main challenge identified was system reliability and real-time response.

Interpretation of Survey 1

The first survey showed that the automated parking idea has practical value, but the concept needed clearer explanation and reliable implementation. Since the expression bot idea was not implemented, it was excluded from the final project scope. The feedback helped the team focus on system stability, real-time response, and proper hardware-software integration for the Automated Car Parking System.

E Survey 2: Prototype Feedback

The second survey was conducted after presenting the Automated Car Parking System concept to a respondent from a garage-related professional background. The purpose was to evaluate product understanding, usefulness, purchase interest, price acceptability, and possible improvement areas.

Survey Questions

1. What is your job role?
2. Have you understood the product concept?
3. Do you think the product will be helpful for your workplace?
4. Would you purchase this product?
5. Do you think the product price is acceptable?
6. How much money would you spend for this type of product?
7. Which part of this product needs improvement?
8. Do you have any additional comments?

Survey Response Summary

Table 3: Summary of Survey 2 Response

Question / Area	**Response Summary**
Respondent Role	Garage Operations Manager.
Understanding of Concept	Rated 5 out of 5.
Workplace Helpfulness	Rated 4 out of 5.
Purchase Interest	Rated 4 out of 5.
Price Acceptability	Rated 4 out of 5.
Expected Budget	The respondent did not provide a clear budget estimate.
Improvement Suggestion	The system should indicate which numbered parking slot the car should go to.
Additional Comment	The respondent stated that the system is good and that modern technology is needed to provide better customer service.

Interpretation of Survey 2

The second survey showed more positive feedback about the Automated Car Parking System. The respondent understood the product concept clearly and considered it useful for a garage or parking-related workplace. The purchase interest and price acceptability were also positive. The main improvement suggestion was to add individual slot guidance so that the driver can know the exact parking slot number. This feature was not implemented in the current prototype but can be added in future work using individual slot sensors and a display or mobile application.

F Evidences of CEP

- **P1: Depth of Knowledge** – Required knowledge of embedded systems, robotics, sensors, RFID, servo control, and IoT communication.
- **P2: Conflicting Requirements** – Balanced low cost, reliable operation, smooth robotic arm movement, real-time display, and online data transmission.
- **P3: Depth of Analysis** – Included analysis of sensor behavior, RFID validation, slot counting, Wi-Fi data transmission, and robotic arm movement.
- **P4: Familiarity of Issues** – Parking automation is a known problem, but using a two-link robotic arm gate created project-specific challenges.
- **P5: Applicable Standards** – Followed common Arduino hardware interfacing practices and used standard libraries.
- **P6: Stakeholder Needs** – Considered drivers, parking managers, and system maintainers.
- **P7: Interdependence** – Sensors, RFID, servo motor, LCD, Wi-Fi module, and online server were interconnected.

G Evidences of CEA

- **A1: Range of Resources** – Used Arduino UNO, IR sensors, RFID module, ESP-01S, LCD, servo motor, robotic arm materials, Arduino IDE, and online database service.
- **A2: Level of Interaction** – Multiple hardware and software modules interacted to complete entry and exit operations.

- **A3: Innovation** – Used a two-link robotic arm gate mechanism instead of a simple fixed barrier gate.
- **A4: Consequences for Society and Environment** – Can reduce waiting time, improve parking management, and support efficient use of parking spaces.
- **A5: Familiarity** – Used familiar technologies but combined them in a practical robotics and IoT-based parking automation system.